

Project Design Phase

Solution Architecture

Date: 30 September 2026

Team ID: SWTID-2026-9911

Project Name: Auto Ticket Classification using Flow Designer

Team Size: 4

Team Leader: Indhu T

Team Members: Mahalakshmi M, Venmathi S, Amala Prathiba J

Maximum Marks: 4 Marks

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SOLUTION ARCHITECTURE

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Solution Architecture Overview:

The solution architecture for "Auto Ticket Classification using Flow Designer" defines how ServiceNow automates and manages the school IT helpdesk incident process. The solution connects student and teacher incident requests, keyword extraction, automated categorization, IT assignment workflows, real-time notifications, and status dashboards into a single, controlled process.

Key Objectives:

- Provide a simple ServiceNow incident form for students and teachers to submit IT requests (Wi-Fi, projector, password, slow computers).
- Automatically analyze short description and description fields using keyword matching logic upon ticket submission.
- Route tickets instantly through Flow Designer to the appropriate IT Support assignment group.

- Automatically set ticket priority, category, and subcategory without manual intervention.
- Send automated status notifications to the requester and assigned IT staff at key resolution stages.
- Maintain an auditable log of ticket creation, auto-classification, updates, and closure history.

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SOLUTION ARCHITECTURE DIAGRAM (DATA FLOW)

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[Requester]

(Student / Teacher)

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| Submits Incident Form

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[ServiceNow Portal]

(Incident Request Form: Short Desc & Desc)

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| Validates & Triggers

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[Workflow & Rules (Flow Designer)]

(Keyword Analysis: Wi-Fi, Projector, Password, Slow PC)

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|-----|

| Categorized

| Record Details

| Status Update

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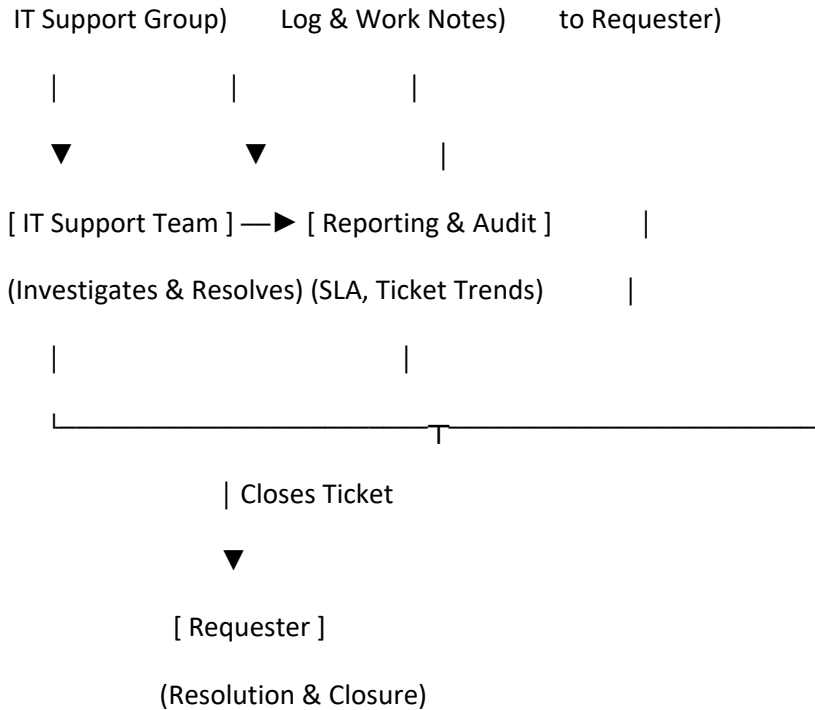
[Auto Assignment]

[ServiceNow Records] [Notifications]

(Routes to Specific

(Incident Table, Audit

(Email / Portal Updates



Key Flow Summary:

Requester → ServiceNow Portal Form → Flow Designer Keyword Analysis → Automatic Classification & Routing → IT Support Resolution → Notification → Closure & Reporting. All activities are recorded in ServiceNow for tracking and reporting.

TECHNOLOGY / SERVICENOW COMPONENTS

Component	Purpose
Service Catalog / Request Form	Collects incident details (User ID, Short Description, Detailed Description) from requesters.

Flow Designer / Workflow | Automates keyword parsing, category/subcategory tagging, task routing, and status transitions.

Business Rules / Data Policies | Enforces required fields (e.g., Short Description cannot be blank) and triggers classification rules.

Incident Management Table | Stores ticket records, assigned categories, routing details, and resolution history.

Notifications | Sends real-time email/portal updates upon creation, classification, assignment, and closure.

Reports / Dashboards | Monitors ticket volume by category, classification accuracy, IT resolution time, and SLAs.

Reference: ServiceNow School IT Helpdesk Use Case - Solution Architecture prepared for Project Design Phase.